Institute of Technological Studies of Bizerte

ELNI (L3-S5) — Dept. of EE	**AY: 2026-2027**
**Midterm Exam	ROS 2**	**Oct 2026**
Teacher: A. Mhamdi	**Time Limit: 60 min**

This exam consists of **11** pages. Please check that it is complete. The **2** tasks are independent and may be solved in any order. The following rules apply:

① **One handwritten, double-sided A4 sheet** is allowed.
② **No electronic material** may be used, except for a basic calculator.
③ **Unjustified or unsupported answers** will not receive full credit.
④ **Round all numerical results** to three decimal places *(i.e., to the nearest thousandth).*
⑤ **Correct answers** receive the points indicated. There is no negative marking.

Task №1 — Drawing a Triangle with Turtlesim

⏳20 min | (5 points)

SOLUTIONS

The objective of this exercise is to control a turtle in the Turtlesim simulator to draw an equilateral triangle using **ROS 2** topics and services. The triangle has the following characteristics:

- Side length: $L = 3$ units.
- Initial position: $(x_0, y_0) = (5, 5)$.
- Initial orientation: $\theta_0 = 0$ rad.
- Linear velocity: $v = 1$ unit/s.
- Angular velocity: $\omega = \frac{\pi}{3}$ rad/s.

The turtle must return to its initial position and orientation after completing the drawing.

(a) (0.25 pt) Give the **ROS 2** command to launch the Turtlesim simulator.

```shell
ros2 run turtlesim turtlesim_node
```

(b) (0.25 pt) Give the command to list all available **ROS 2** topics. Identify the topic used to control the turtle's movement.

List all available topics:

```shell
ros2 topic list
```

The topic used to control the turtle is `/turtle1/cmd_vel`.

(c) (0.5 pt) Before drawing the triangle, the turtle must be positioned at (5, 5) with orientation 0 rad. Identify the service that can perform this operation and write the command to call it.

The service is `/turtle1/teleport_absolute` of type `turtlesim/srv/TeleportAbsolute`.

Command to call the service:

```shell
ros2 service call /turtle1/teleport_absolute turtlesim/srv/TeleportAbsolute "{x: 5.0, y: 5.0, theta: 0.0}"
```

(d) (0.5 pt) Determine the message type associated with `/turtle1/cmd_vel`. Then display its structure. Which fields control the turtle's linear and angular velocities?

Identify the message type:

```shell
ros2 topic info /turtle1/cmd_vel
```

Message type: `geometry_msgs/msg/Twist`.

Display its structure:

```shell
ros2 interface show geometry_msgs/msg/Twist
```

- **linear.x** Forward linear velocity.
- **angular.z** Angular velocity around the *z*-axis.

(e) (0.75 pt) Determine:
- The angle through which the turtle must rotate at each vertex.
- The duration of each straight movement.
- The duration of each rotation.

The exterior angle of a regular polygon with n sides is $2\frac{\pi}{n}$. Use $t = \frac{d}{v}$ and $t = \frac{\theta}{\omega}$.

- The exterior angle of an equilateral triangle is: $\theta = \frac{2\pi}{3} = 120°$
- The duration of a straight movement is: $t_m = \frac{L}{v} = \frac{3}{1} = 3\ \text{s}$
- The duration of each rotation is: $t_r = \frac{\theta}{\omega} = \frac{\frac{2\pi}{3}}{\frac{\pi}{3}} = 2\ \text{s}$
- Total theoretical duration: $T = 3(t_m + t_r) = 3(3 + 2) = 15\ \text{s}$

(f) (1 pt) Write the **ROS 2** commands necessary to:
- Make the turtle move forward.
- Make the turtle rotate counterclockwise.

Specify the velocity values and the required duration for each movement.

Move forward Publish for approximately 3 seconds.

```
ros2 topic pub -t 3 -r 1 /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 1.0}, angular: {z: 0.0}}"
```
Shell

Rotate counterclockwise Publish for approximately 2 seconds.

```
ros2 topic pub -t 2 -r 1 /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0}, angular: {z: 1.0472}}"
```
Shell

✦ *We need to repeat the forward movement and rotation three times to complete the triangle. **ROS 2** communication delays may introduce positioning errors. The resulting triangle is approximate, not geometrically exact.*

(g) (0.75 pt) After drawing the triangle, determine the **ROS 2** topic that provides the turtle's current pose. Explain how to verify that the turtle has returned to its initial pose.

The turtle's pose is published on `/turtle1/pose`.

```
ros2 topic echo /turtle1/pose
```
Shell

The final pose should approximately satisfy: $x = 5, y = 5$ and $\theta = 0$ rad. The final orientation may also be represented by an angle equivalent to 0 modulo 2π. Small differences may occur due to timing and numerical errors.

(h) (1 pt) During the execution, the turtle does not return exactly to its initial position after completing the triangle. Propose a method to correct this positioning error **without using the teleportation service**.

Explain how the information published on `/turtle1/pose` can be used to determine the necessary corrective movements.

i *Consider the difference between the current and target positions. The turtle can first rotate toward the target, then move forward by the remaining distance.*

First, obtain the turtle's current pose:

```
ros2 topic echo /turtle1/pose --once
```
Shell

Let (x, y, θ) be the current pose and $(x_0, y_0) = (5, 5)$ the target position.

- Calculate the positioning error: $\Delta x = x_0 - x$ and $\Delta y = y_0 - y$.
- The required distance is: $d = \sqrt{(\Delta x)^2 + (\Delta y)^2}$.

- The direction toward the target is: $\theta_d = \text{atan2}(\Delta y, \Delta x)$.
- The required rotation is: $\Delta\theta = \theta_d - \theta$.
- Normalize $\Delta\theta$ to $[-\pi, \pi]$.

Correction procedure:

1. Rotate by $\Delta\theta$ using `angular.z`.
2. Move forward by distance d using `linear.x`.
3. Stop the turtle.
4. Read `/turtle1/pose` again to verify the result.
5. Correct the final orientation to $\theta_0 = 0$ rad.

With constant velocity magnitudes, the durations are $t_r = \dfrac{|\Delta\theta|}{|\omega|}$ and $t_m = \dfrac{d}{v}$.

✦ *Small residual errors can be reduced by repeating the correction procedure.*

SOLUTIONS

Appendix: Turtlesim Quick Reference

Starting turtlesim

Open a terminal and start the simulator window:

```shell
ros2 run turtlesim turtlesim_node
```

In a second terminal, start the keyboard controller:

```shell
ros2 run turtlesim turtle_teleop_key
```

Inspecting the system

Command	Description
`ros2 node list`	Lists the running nodes
`ros2 node info /turtlesim`	Shows the topics and services of a node
`ros2 topic list`	Lists the active topics
`ros2 topic echo /turtle1/pose`	Prints the turtle position messages
`ros2 topic info /turtle1/cmd_vel`	Shows the message type of a topic
`ros2 service list`	Lists the available services
`rqt_graph`	Shows nodes and topics as a graph

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Publishing to a topic

Move the turtle without the keyboard by publishing a velocity message once:

```shell
ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist
"{linear: {x: 2.0}, angular: {z: 1.8}}"
```

Calling services

Clear the drawn path:

```shell
ros2 service call /clear std_srvs/srv/Empty
```

Teleport to an absolute position *(x, y, and orientation in radians)*:

```shell
ros2 service call /turtle1/teleport_absolute turtlesim/srv/
TeleportAbsolute "{x: 5.0, y: 5.0, theta: 0.0}"
```

Teleport relative to the current position *(move forward by* `linear`*, then rotate by* `angular` *radians)*:

```shell
ros2 service call /turtle1/teleport_relative turtlesim/srv/
TeleportRelative "{linear: 2.0, angular: 1.57}"
```

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TASK №2 ⏳40 min | (15 points)

(a) (0.5 pt) Which command gives the file owner permission to execute `script.sh`?

- ○ `chmod -r script.sh`
- ○ `chown +x script.sh`
- ○ `exec script.sh`
- ✓ `chmod u+x script.sh`

(b) (0.5 pt) Which one is a valid function declaration in Bash?

- ✓ `greet() { echo "Hi"; }`
- ○ `def greet():`
- ○ `fn greet {}`
- ○ `function greet[]`

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(c) (0.5 pt) What does the pipe operator `|` do in `ls | grep txt`?

- ○ Saves the output to a file
- ○ Runs both commands in the background
- ✓ Sends the output of `ls` as input to `grep`
- ○ Runs `grep` only if `ls` fails

(d) (0.5 pt) What does the `-i` option do in `grep -i "error" log.txt`?

- ○ Inverts the match
- ✓ Ignores upper/lower case
- ○ Counts matching lines
- ○ Shows line numbers

DO NOT WRITE ANYTHING HERE

✂ ----------

(e) (0.5 pt) What is the name of the first line `#! /bin/bash` in a shell script?

○ Comment ✓ Shebang ○ Pipe ○ Alias

(f) (0.5 pt) Inside a script, which variable holds the first argument passed to it?

✓ `$1` ○ `$0` ○ `$#` ○ `$?`

(g) (0.5 pt) Which line correctly assigns a value to a variable in Bash?

○ `NAME = "Rami"` ○ `$NAME="Rami"` ○ `set NAME "Rami"` ✓ `NAME="Rami"`

(h) (0.5 pt) What does `echo "done" >> log.txt` do?

○ Overwrites `log.txt`

○ Deletes `log.txt`

✓ Appends “done” to `log.txt`

○ Reads `log.txt`

(i) (0.5 pt) What does the special variable `$?` contain?

○ The script’s name

✓ Exit status of the last command

○ Number of arguments

○ The current process ID

(j) (0.5 pt) What permissions does `chmod 755 script.sh` set?

○ Full access for everyone

○ Read-only for everyone

○ Owner r-x, others rwx

✓ Owner rwx, group/others r-x

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(k) (0.5 pt) In `tmux`, what is the default prefix key combination?

○ `Ctrl+a` ✓ `Ctrl+b` ○ `Ctrl+t` ○ `Ctrl+x`

(l) (0.5 pt) In `tmux`, after pressing the prefix, which key splits the current pane into left and right panes?

○ `c` ○ `"` ○ `d` ✓ `%`

(m) (0.5 pt) Which crontab entry runs `/home/user/backup.sh` every day at 2:30 AM?

○ `2 30 * * * /home/user/backup.sh`

○ `* * 2 30 * /home/user/backup.sh`

✓ `30 2 * * * /home/user/backup.sh`

○ `2:30 * * * * /home/user/backup.sh`

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> i *A cron schedule has five fields, in this order: minute, hour, day of month, month, day of week. A `*` means "every" value. `*/N` in a field means "every N units" of that field. Check which field it is in.*

(n) (0.5 pt) What does the crontab schedule `*/5 * * * *` mean?

○ Every 5 hours

✓ Every 5 minutes

○ At minute 5 of every hour

○ Every 5 days

(o) (0.5 pt) Which data type stores key-value pairs in Python?

✓ `dict` ○ `list` ○ `tuple` ○ `set`

(p) (0.5 pt) Which function is used to display text on the screen in Python?

○ `echo()` ○ `display()` ✓ `print()` ○ `write()`

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(q) (0.5 pt) Which line correctly creates a list in Python?

- ○ `items = (1, 2, 3)`
- ✓ `items = [1, 2, 3]`
- ○ `items = {1, 2, 3}`
- ○ `items = <1, 2, 3>`

(r) (0.5 pt) What is the result of `type(3.14)`?

- ○ `<class 'int'>`
- ○ `<class 'str'>`
- ○ `<class 'decimal'>`
- ✓ `<class 'float'>`

(s) (0.5 pt) Which one is a valid function definition in Python?

- ✓ `def greet(name):`
- ○ `function greet(name):`
- ○ `greet(name) = def:`
- ○ `func greet(name) {}`

(t) (0.5 pt) What does `len("hello")` return?

○ 4 ○ 6 ✓ 5 ○ "hello"

(u) (0.5 pt) What does `[1, 2, 3].append(4)` do?

- ○ Creates a new list `[1, 2, 3]` and discards `4`
- ○ Inserts `4` at the beginning of the list
- ✓ Adds `4` to the end of the list
- ○ Replaces the last element with `4`

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(v) (0.5 pt) What is the output of the following code? `x = "hello"; print(x[1])`

○ h ✓ e ○ l ○ o

(w) (0.5 pt) Which command lists all currently running **ROS 2** nodes?

○ `ros2 list nodes`

✓ `ros2 node list`

○ `ros2 show nodes`

○ `ros2 topic nodes`

(x) (0.5 pt) Which command starts the turtlesim window?

○ `ros2 launch turtlesim turtle`

○ `ros2 start turtlesim turtlesim_node`

✓ `ros2 run turtlesim turtlesim_node`

○ `ros2 turtlesim run`

(y) (0.5 pt) Which command lets you drive the turtle with the keyboard arrows?

✓ `ros2 run turtlesim turtle_teleop_key`

○ `ros2 run turtlesim keyboard_node`

○ `ros2 run turtlesim move_turtle`

○ `ros2 topic keyboard /turtle1`

(z) (0.5 pt) What does `ros2 topic list` display?

○ The installed **ROS 2** packages

○ The running nodes

○ The available services

✓ The currently active topics

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✂ ----------

(aa) (0.5 pt) Which command prints the messages published on the topic `/turtle1/cmd_vel`?

- ○ `ros2 topic print /turtle1/cmd_vel`
- ✓ `ros2 topic echo /turtle1/cmd_vel`
- ○ `ros2 topic read /turtle1/cmd_vel`
- ○ `ros2 node echo /turtle1/cmd_vel`

(ab) (0.5 pt) Which statement best describes a **ROS 2** service?

- ○ A continuous one-way stream of data from a publisher to subscribers
- ○ A file that stores node parameters
- ✓ A request/response communication between a client and a server
- ○ A tool that starts several nodes at once

(ac) (0.5 pt) What does `rqt_graph` show?

- ✓ A graph of the running nodes and the topics connecting them
- ○ A plot of the turtle's position over time
- ○ A list of all installed **ROS 2** packages
- ○ The CPU usage of each node

(ad) (0.5 pt) Which command lists all the services currently available?

- ○ `ros2 topic services`
- ○ `ros2 node services`
- ○ `ros2 list service`
- ✓ `ros2 service list`

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